Neural Empirical Mode Decomposition: Differentiable Signal Decomposition via Adaptive Frequency-Domain Partition with Provable Architectural Guarantees

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***Abstract*—Empirical mode decomposition (EMD) and variational mode decomposition (VMD) are widely used in biomedical, industrial, and geophysical signal processing but are not differentiable and cannot be jointly trained with a downstream task. Learned alternatives either give up interpretability or rely on soft physics losses that admit degenerate solutions. We introduce Neural Empirical Mode Decomposition (N-EMD), a differentiable signal-adaptive filter bank whose architecture guarantees three properties by construction: exact reconstruction of the input, a Parseval–Cauchy–Schwarz upper bound on total component energy, and non-negative bandpass filter responses. Pairwise orthogonality is a corollary that becomes exact as the softmax temperature tends to zero. A neural analyzer maps the input spectrum to K log-responses, a softmax over the K dimension produces a partition of unity, and the inverse FFT of each filtered spectrum yields an IMF. All three guarantees are structural consequences of this parameterisation and hold at initialisation, during training, and on unseen inputs. We document a failure mode of the more obvious approach — training a generic sifting network with seven soft physics losses kept finding a new degenerate minimum whenever an additional loss was added. On a three-class classification benchmark (5 seeds, 3 SNR levels), end-to-end N-EMD matches a capacity-matched VMD+MLP baseline at SNR 3 dB (94.2 ± 2.9 vs. 92.7 ± 0.8 percent) and loses at 10 and 20 dB with high training-run variance, so N-EMD's advantage is narrow and concentrated in the low-SNR regime; the architectural guarantees transfer unchanged to held-out distributions and to real pupillometry data, where neither N-EMD nor any baseline (including fixed VMD and a raw CNN) beats majority-class prediction — a dataset-level rather than method-level null.**

***Index Terms*—Signal decomposition, empirical mode decomposition, variational mode decomposition, differentiable signal processing, partition of unity, task-aware representation learning, filter banks.**

I. Introduction

SIGNAL decomposition is a foundational tool for the analysis of nonstationary data. A signal $s(t)$ is written as a sum of oscillatory components plus a slow residual, and each component is treated as a narrow-band carrier of amplitude, frequency, or phase information on its own. Empirical mode decomposition (EMD) [1] was introduced for this purpose nearly three decades ago and has since anchored a family of related methods: ensemble EMD [2], CEEMDAN [3], variational mode decomposition (VMD) [4], empirical wavelet transform [5], and synchrosqueezed wavelets [6]. These methods are widely used in bearing-fault diagnosis [7], biomedical signal analysis [8], seismology, and many other settings because the components they produce are interpretable — each one occupies a well-defined frequency band and can be related back to physical phenomena.

Three limitations hold the classical family back. First, the methods are not differentiable: they use sifting (EMD) or alternating-direction optimisation via ADMM (VMD) [9]. This rules out joint training with any downstream model, so the decomposition and the task it serves cannot share gradients. Second, the decomposition is chosen once per signal by the method's built-in heuristic, with no mechanism for adapting to what a downstream task actually needs. Third, the properties one typically wants — exact reconstruction, orthogonality, energy conservation — are approximate empirical outcomes rather than guarantees. On the canonical three-component signals used throughout this paper, EMD's orthogonality index ranges from 0.04 to 0.10 depending on input, and VMD's energy ratio can drop to 0.60 on noisy signals.

Learned alternatives face a trade-off. End-to-end 1-D convolutional networks give up interpretability entirely: the decomposition is implicit inside the network. Physics-constrained learners trained with soft losses for reconstruction, orthogonality, and bandwidth suffer a subtler problem: soft constraints penalise deviation from a property but do not forbid it, so a generic function approximator can satisfy every physics loss while still producing degenerate components. Section IV records our own experience with this failure mode: after six loss terms and two months of tuning, we were still meeting each fix with a new degenerate minimum.

We take a different approach. Rather than imposing the desired properties through losses, we encode them in the architecture. Concretely, Neural Empirical Mode Decomposition (N-EMD) is a signal-adaptive filter bank in which a small analyzer network maps the input spectrum $S(f)$ to K real-valued log-responses; a softmax over the K dimension turns them into a partition of unity — the K filters sum to 1 at every frequency bin. Applying each filter in the frequency domain and inverse-transforming yields IMFs whose sum reconstructs the input exactly, whose total energy is bounded above by the signal's (Cauchy–Schwarz), and whose filter responses are non-negative by construction. No loss term is needed for any of these properties; they follow from the algebra of the softmax.

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Contributions.

1) *A partition-of-unity architecture for signal decomposition.* We introduce a neural adaptive filter bank whose softmax parameterisation provides exact reconstruction, a Parseval bound on component energy, non-negativity of filter responses, and arbitrarily-close orthogonality at low temperature (Propositions 1–3 and Corollary 1). All four properties hold on any input, at initialisation, during training, and at inference.
2) *End-to-end task-aware training.* The architecture is differentiable; task gradients can shape the filter bank. Section V-D shows that an N-EMD classifier trained end-to-end at SNR 3 dB improves over a fixed-VMD baseline by $+7.8$ pp and over a fixed-EMD baseline by $+14.0$ pp, while retaining a physically-interpretable decomposition.
3) *A negative result on soft physics constraints.* We document four progressive failure modes of a high-capacity sifting network trained with seven soft physics losses (Section IV). Each failure motivated an additional loss term; each additional term exposed a new degenerate minimum. We argue that this pattern is structural: soft constraints do not shrink the feasible set of a non-convex optimisation.
4) *Honest SNR-regime analysis.* The end-to-end advantage inverts at high SNR: at 20 dB, fixed VMD+MLP outperforms N-EMD by 17.5 pp (Section V-D). Task-aware learned decomposition is most valuable in the noisy regime where fixed preprocessing degrades most, and least valuable in clean regimes where good features are nearly given.

Paper structure. Section II surveys classical decomposition, learned decomposition, and differentiable DSP. Section III defines the N-EMD architecture, proves its three guarantees, and specifies the training losses and procedure. Section IV documents the negative result that motivates the architectural approach. Section V reports experiments on decomposition quality, generalisation, task-aware classification across three SNR levels, and biomedical application to pupillometry. Section VI discusses limitations; Section VII concludes.

II. Background and Related Work

A. Classical Signal Decomposition

EMD [1] decomposes a signal by iterative sifting: it identifies local extrema, fits cubic-spline upper and lower envelopes, subtracts their mean, and repeats until each IMF satisfies a narrow-band condition. EMD has well-known problems — mode mixing, boundary effects, and sensitivity to noise [10]. These problems motivated the noise-assisted variants EEMD [2] and CEEMDAN [3], which average over an ensemble of noise realisations at considerable computational cost. Flandrin et al. [11] showed that EMD acts as a dyadic filter bank on stochastic input, which suggests the filter-bank framing we make explicit in this paper.

VMD [4] reformulates decomposition as a constrained optimisation: find K modes whose sum is the input and whose individual bandwidths are minimised. Its K centre frequencies are optimised per signal by ADMM [9], which gives VMD narrow-band components but forces a fresh optimisation for every new input. Empirical wavelet transform [5] partitions the Fourier axis into adaptive bands by a peak-detection heuristic; synchrosqueezing [6] refines a time-frequency ridge and then reconstructs components by integration. All of these methods produce interpretable decompositions but are not differentiable and not task-adaptive.

B. Learned Filter Banks and Differentiable Frontends

Several threads of prior work produce interpretable, differentiable filter banks; N-EMD is positioned relative to each.

SincNet [12] parameterises the first layer of a speech-recognition model as a bank of sinc-windowed bandpass filters, each specified by two learnable cutoffs. The filters are interpretable and differentiable, but the bank does not satisfy any partition-of-unity identity; applying the full bank to a signal does not reconstruct it. *LEAF* [13] extends this idea with parametric Gabor filters, per-band compression, and learnable downsampling; it is tuned for classification rather than decomposition and also does not guarantee reconstruction. *Pariente et al.* [14] construct a fully-learnable analysis–synthesis filter bank for speech separation, with an explicit synthesis stage that approximately inverts the analysis; reconstruction is empirical, not a structural property. *Scattering networks* [15], [16] apply cascaded fixed wavelet transforms with provable stability-to-deformation bounds; hybrid learnable variants exist [17]. Scattering is not a decomposition in the EMD/VMD sense — components are not signals of the same length as the input — and reconstruction from scattering coefficients is itself a non-trivial inverse problem. *Learnable wavelet packets* [18] use the dyadic tree of classical wavelets with learnable filter taps; reconstruction holds under the perfect-reconstruction (PR) conditions from multirate signal processing [19], [20], but the partition is dyadic rather than data-adaptive.

N-EMD differs from all of the above in one specific way: the filter responses are produced by a neural analyzer conditioned on the *current input* (not just by learned parameters shared across inputs), and a softmax across the filter axis enforces $\sum_k H_k(f) = 1$ at every frequency bin. This combination yields exact reconstruction by identity (not by a PR condition on the taps) and a Cauchy–Schwarz bound on total component energy that holds for any input. The closest existing mechanism is the softmax gating of mixture-of-experts models [21], transposed from expert routing to per-frequency filter weighting.

C. Learned Signal Decomposition

Attempts to make EMD differentiable or to learn decomposition directly have been comparatively sparse. Where they exist, they tend to relax one of the desiderata: either the decomposition is implicit inside a black-box network (no bandpass structure that can be plotted), or the reconstruction identity is approximate (a hidden residual absorbs the error), or the decomposition is fixed once for the training distribution rather than adapted per input. N-EMD's specific

claim is to produce exact reconstruction, per-frequency non-negative bandpass weights summing to one, input-adaptive filter responses, and differentiability jointly. We have not found a prior method with this exact set of properties, though the individual ingredients (learnable bandpass filters, softmax partitions, adaptive-per-input networks) each appear in the literature above.

D. Differentiable Signal Processing

The differentiable DSP line of work [22] incorporates explicit signal-processing structure — oscillators, filters, noise generators — into neural networks to improve interpretability and sample efficiency over end-to-end black boxes. N-EMD sits in this same tradition: the forward pass is an rFFT, a convolutional analyzer, a softmax, and an irFFT, with no unexplained operations.

E. Physics-Informed Machine Learning

Physics-informed neural networks [23] and related methods impose physical laws as soft penalties on the training loss. This works well in settings where the physical constraint uniquely determines the solution up to smoothness and boundary conditions. In settings where the physical constraints are *necessary but not sufficient* — as in signal decomposition, where orthogonality, bandwidth, and energy conservation together still admit degenerate solutions — soft penalties are not enough. Section IV documents that failure in detail. The lesson of this paper is complementary to the PINN literature: when the feasible set is not single-valued under the constraints, reparameterise so that the constraints hold by construction.

III. The N-EMD Method

This section presents Neural Empirical Mode Decomposition (N-EMD). The method is a frequency-domain adaptive filter bank trained end-to-end. Three architectural guarantees follow from the construction itself, without any loss term enforcing them: exact reconstruction, a Parseval bound on component energy, and non-negativity of filter responses. Section III-A fixes notation; Section III-B describes the architecture; Section III-C states the guarantees as propositions; Sections III-D and III-E describe the training losses and procedure; Section III-F describes the (permutation-invariant) IMF ordering applied at inference time.

A. Problem Formulation

Let $s \in \mathbb{R}^T$ be a discrete real signal of length T, sampled at rate f_s Hz with Nyquist frequency $f_\mathrm{N} = f_s/2$. Write its one-sided discrete Fourier transform as $S = \mathrm{rFFT}(s) \in \mathbb{C}^N$ with $N = \lfloor T/2 \rfloor + 1$ frequency bins $f_n = n\, f_s/T$ for $n = 0, \ldots, N-1$.

We seek a decomposition

$$s \;=\; \sum_{k=1}^{K} c_k \tag{1}$$

into K real-valued bandpass components $c_k \in \mathbb{R}^T$ (intrinsic mode functions, or IMFs), with the following properties:

1) *Reconstruction*: equality in (1) holds exactly (to numerical precision).
2) *Bandpass concentration*: each c_k is concentrated in a distinct frequency region, with narrow spectral support.
3) *Orthogonality*: the components are pairwise approximately orthogonal in $\ell^2(\mathbb{R}^T)$.
4) *Energy boundedness*: $\sum_k \|c_k\|_2^2$ does not exceed $\|s\|_2^2$, so components cannot inflate and cancel.

Classical EMD satisfies (1) only as an empirical outcome (a residual r_K is typically retained) and provides (2)–(4) only approximately. VMD's reconstruction error is typically of order 10^{-2} and its component energy varies between 0.6 and $1.0 \times \|s\|_2^2$ depending on input; see Section V. We want all four properties as structural consequences of the construction, not as soft targets of a loss function.

B. Partition-of-Unity Filter-Bank Architecture

The core idea is to parameterise the decomposition as a softmax partition of the frequency axis. A single neural network, the *signal analyzer*, maps the input spectrum to K real-valued log-responses; a softmax over the K dimension turns them into per-frequency filter weights that sum to unity at every bin. Each filter is applied in the frequency domain and inverse-transformed. The architecture is illustrated in Fig. 1.

a) Signal analyzer.: The analyzer f_θ is a small one-dimensional convolutional network that operates on the N frequency bins as spatial positions. Its input is a two-channel spectrogram of the magnitude and phase of S,

$$\big[|S(f)|,\ \angle S(f)\big] \in \mathbb{R}^{2\times N}, \tag{2}$$

and its output is a stack of K real-valued log-responses $s_k(f) \in \mathbb{R}^{K\times N}$:

$$\big(s_1(f), \ldots, s_K(f)\big) \;=\; f_\theta\big(|S|,\ \angle S\big). \tag{3}$$

The default configuration has an input 1×7 convolution lifting the two channels to $d = 64$ hidden channels, three residual blocks of two 1×5 convolutions with batch normalisation and GELU activations, and a 1×1 output convolution projecting to K channels. The total parameter count is 125,315 for $K = 3$.

b) Frequency partition.: Per-frequency filter responses are obtained from a temperature-scaled softmax across the K dimension,

$$H_k(f) \;=\; \frac{\exp\big(s_k(f)/\tau\big)}{\sum_{j=1}^{K} \exp\big(s_j(f)/\tau\big)}, \qquad \tau > 0, \tag{4}$$

so that at every bin f the K filter weights are non-negative and sum to one. The temperature τ controls filter sharpness: as $\tau \to 0^+$, each $H_k(f)$ tends to an indicator of the highest log-response, producing a disjoint partition of the frequency axis; large τ gives broader, smoother filters. We anneal τ during training (Section III-E).

c) IMF extraction.: The K IMFs are obtained by point-wise multiplication of the filters with the complex spectrum, followed by the inverse real FFT at length T:

$$c_k \;=\; \mathrm{irFFT}_T\big(H_k(f)\, S(f)\big). \tag{5}$$

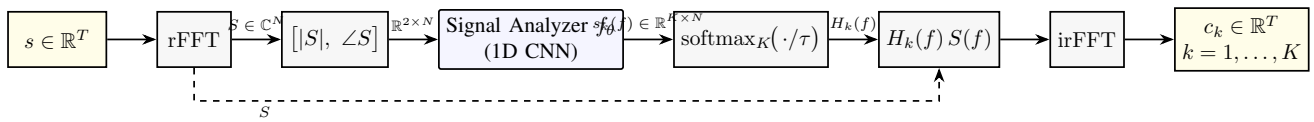

Fig. 1. Architecture of N-EMD. The input spectrum is fed to a 1D CNN analyzer that predicts K log-filter responses. A softmax over the K dimension enforces the partition-of-unity condition $\sum_k H_k(f) = 1$. Each filter is multiplied with the input spectrum and inverse-transformed to yield an IMF. The dashed path reuses the original complex spectrum S at the pointwise-multiply stage.

The entire pipeline is differentiable with respect to the analyzer parameters θ, so the decomposition can be trained jointly with any downstream differentiable loss (Section V-D).

Remark 1 (Length-agnostic operation). The analyzer is a fully-convolutional 1D network over frequency bins; no fixed length is baked into its weights. A model trained at signal length T can be applied at any other length T' (and any sample rate) with no modification, since $|S|$ and $\angle S$ are the only inputs. The trained checkpoint used throughout the paper was optimised at $T = 512$, $f_s = 512$ Hz and is applied at $T = 1024$, $f_s = 1000$ Hz without retraining.

Remark 2 (Phase channel). Including the phase $\angle S$ alongside $|S|$ lets the analyzer distinguish signals with identical power spectra but different temporal structure (e.g., a steady tone vs. a matched-energy chirp). A magnitude-only ablation is a natural future check.

Remark 3 (On the name *N-EMD*). The mechanism here is a one-shot global filter bank, not EMD's iterative envelope sifting. We retain the EMD framing because (i) EMD was itself characterised as an adaptive filter bank on stochastic input [11], so the interpretive role of an IMF as a narrow-band oscillatory component is the role our filters produce, and (ii) the output shape (ordered bandpass components) is the one classical-EMD users already know how to read. Readers who prefer a mechanism-level name may read the contribution as a *neural adaptive filter bank with exact reconstruction*; nothing in the theory or experiments depends on the label.

C. Architectural Guarantees

We now state the three properties that hold for N-EMD by construction, independent of any loss or training procedure. All three follow from a single algebraic fact — that the softmax output in (4) is a non-negative partition of unity over K elements — combined with the linearity of the inverse FFT and Parseval's identity.

Proposition 1 (Exact reconstruction)**.** *For any input signal $s \in \mathbb{R}^T$, any temperature $\tau > 0$, and any analyzer output $s_k(f) \in \mathbb{R}^{K\times N}$, the components defined by* (5) *satisfy*

$$\sum_{k=1}^{K} c_k(t) \;=\; s(t), \qquad t = 0, 1, \ldots, T-1, \tag{6}$$

with equality exact up to the floating-point round-off of the rFFT / irFFT *pair.*

Proof. By linearity of the inverse FFT and (4),

$$\sum_{k=1}^{K} c_k = \mathrm{irFFT}_T\Big(\Big(\sum_{k=1}^{K} H_k(f)\Big) S(f)\Big) = \mathrm{irFFT}_T(S(f)) = s,$$

where the penultimate equality uses $\sum_k H_k(f) = 1$ at every f and the last uses that irFFT_T inverts rFFT on real signals of length T. □

Proposition 2 (Energy boundedness)**.** *For any input s,*

$$\sum_{k=1}^{K} \|c_k\|_2^2 \;\le\; \|s\|_2^2, \tag{7}$$

with equality if and only if the filters form a disjoint partition, i.e. for each f there is some $k(f)$ with $H_{k(f)}(f) = 1$ and $H_j(f) = 0$ for $j \neq k(f)$.

Proof. By Parseval's identity and (5), $\|c_k\|_2^2$ equals (up to a constant depending on the FFT normalisation) $\sum_f H_k(f)^2 |S(f)|^2$. Summing over k and exchanging the order of summation,

$$\sum_{k=1}^{K} \|c_k\|_2^2 = \sum_f |S(f)|^2 \sum_{k=1}^{K} H_k(f)^2. \tag{8}$$

Since $H_k(f) \in [0,1]$ and $\sum_k H_k(f) = 1$, the Cauchy–Schwarz inequality (equivalently, the power-mean inequality) gives $\sum_k H_k(f)^2 \le \big(\sum_k H_k(f)\big)^2 = 1$, with equality iff the vector $\big(H_1(f), \ldots, H_K(f)\big)$ is a standard basis vector. The result follows from Parseval once more, $\sum_f |S(f)|^2 = \|s\|_2^2$. □

Proposition 3 (Non-negativity and bandpass interpretation)**.** *For any input, any temperature $\tau > 0$, and any (k, f) the filter response satisfies $H_k(f) \ge 0$. Consequently, each IMF c_k is a bandpass-filtered version of s obtained by multiplying the spectrum $S(f)$ by a non-negative, per-bin attenuation $H_k(f) \in [0,1]$.*

Proof. $H_k(f)$ is the k-th component of a softmax output, hence a probability in $[0,1]$; $\sum_k H_k(f) = 1$ was shown in Proposition 1. Equation (5) is the inverse transform of $S(f)$ modulated by $H_k(f) \in [0,1]$, which is the definition of a linear bandpass filter with real, non-negative frequency response. □

Propositions 1–3 are stated without any assumption on the analyzer: they hold at initialisation, during training, and at inference, on any input signal, with any K and any $\tau > 0$. They are the structural counterpart of the soft constraints (reconstruction loss, energy-conservation loss, non-negativity penalty) that physics-informed approaches typically add to the training objective. Because they are enforced by the parameterisation itself, the training loss need not contain any term for them. We return to this point in Section IV.

Corollary 1 (Approximate orthogonality)**.** *For any two IMFs c_k and c_ℓ with $k \neq \ell$,*

$$|\langle c_k, c_\ell \rangle| \;\leq\; \sum_f H_k(f)\, H_\ell(f)\, |S(f)|^2. \tag{9}$$

In particular, if H_k and H_ℓ have disjoint support on the frequency axis, then $\langle c_k, c_\ell \rangle = 0$.

Proof. By Parseval, $\langle c_k, c_\ell \rangle = \sum_f H_k(f) H_\ell(f) |S(f)|^2$ (up to the FFT normalisation), and every summand is non-negative by Proposition 3. Disjoint support makes every summand zero. □

As $\tau \to 0^+$ the softmax partition becomes the argmax indicator; supports become disjoint and Corollary 1 makes the cross terms exactly zero. At finite τ, the bound on overlap is governed by the signal-weighted product of adjacent filters. In Section V we report measured orthogonality indices of 0.00 ± 0.01 across five test distributions, compared with 0.04–0.10 for EMD and VMD.

D. Training Losses

Because Propositions 1–3 relieve the loss of the reconstruction/energy-conservation duties, the training objective reduces to four terms that shape the *quality* of the decomposition rather than its feasibility:

$$\begin{aligned}\mathcal{L} \;&= \lambda_{\text{sharp}}\, \mathcal{L}_{\text{sharp}} + \lambda_{\text{order}}\, \mathcal{L}_{\text{order}} + \lambda_{\text{ortho}}\, \mathcal{L}_{\text{ortho}} \\ &\quad + \lambda_{\text{bal}}\, \mathcal{L}_{\text{bal}} + \lambda_{\text{task}}\, \mathcal{L}_{\text{task}}.\end{aligned} \tag{10}$$

The first four are always present; the task term is optional and only activates for end-to-end training (Section V-D). Default weights are $\lambda_{\text{sharp}} = 2$, $\lambda_{\text{order}} = 2$, $\lambda_{\text{ortho}} = 0.1$, $\lambda_{\text{bal}} = 5$; we use these throughout.

a) Signal-weighted sharpness.: A broad, flat filter $H_k(f) \approx 1/K$ is a valid partition of unity (summing to one with zero entropy contribution per bin) but corresponds to no decomposition at all. We penalise the signal-weighted Shannon entropy of each filter,

$$\mathcal{L}_{\text{sharp}} \;=\; \frac{1}{K \log N} \sum_{k=1}^{K} \mathrm{H}\big(p_k\big), \tag{11}$$

with $p_k(f) = H_k(f)|S(f)|^2 / \sum_{f'} H_k(f')|S(f')|^2$ the signal-weighted filter density. so that filters that isolate a narrow band of *signal* energy are rewarded, but no penalty is applied to broad-but-silent frequency regions. This term is essential: without it, the network satisfies every remaining constraint by choosing $H_k(f) = 1/K$ everywhere.

b) Centroid separation.: For each IMF we compute the signal-weighted spectral centroid $\bar{f}_k$ and normalise it to the Nyquist range, $\hat{f}_k = \bar{f}_k / f_{\text{N}} \in [0, 1]$. The centroid-separation loss combines three terms (ordering, repulsion, coverage):

$$\begin{aligned}\mathcal{L}_{\text{order}} \;&=\; \frac{1}{K-1} \sum_{k=1}^{K-1} \big[\max(0, \hat{f}_{k+1} - \hat{f}_k + m)\big]^2 \\ &- \frac{\eta_r}{K(K-1)} \sum_{i \neq j} \log\big(|\hat{f}_i - \hat{f}_j| + \epsilon\big) - \eta_c \operatorname{Var}\big(\hat{f}_1, \ldots, \hat{f}_K\big).\end{aligned} \tag{12}$$

The first (squared hinge, margin $m = 0.02$) enforces descending centroids; the second is a pairwise log-distance repulsion that acts on all pairs, not only consecutive ones; the third rewards coverage of the Nyquist axis via centroid variance. Each of these terms addresses a distinct failure mode discovered during development (Section IV); the full ablation is in Appendix D. Default sub-weights are $\eta_r = 0.5$, $\eta_c = 0.3$ and $\epsilon = 10^{-4}$.

c) Soft orthogonality.: Although Corollary 1 already gives near-orthogonality at small τ, a mild soft penalty on the cosine similarity between IMF pairs accelerates training:

$$\mathcal{L}_{\text{ortho}} \;=\; \frac{1}{K(K-1)} \sum_{i \neq j} \Big(\tfrac{\langle c_i, c_j \rangle}{\|c_i\|_2\, \|c_j\|_2 + \epsilon} \Big)^2. \tag{13}$$

Its weight $\lambda_{\text{ortho}} = 0.1$ is small; in the trained model the measured orthogonality index is at the 10^{-4} level.

d) Filter balance.: The softmax partition admits a degenerate solution in which one filter covers the whole axis and the remaining $K-1$ are zero; this is a valid partition of unity but a useless decomposition. The balance loss requires each filter to cover at least a minimum fraction of the frequency axis on average:

$$\mathcal{L}_{\text{bal}} \;=\; \frac{1}{K} \sum_{k=1}^{K} \max\big(0,\; \alpha - \overline{H_k}\big), \qquad \overline{H_k} = \frac{1}{N} \sum_f H_k(f), \tag{14}$$

with $\alpha = 1/(2K)$. With $K = 3$, each filter must cover at least one sixth of the axis on average; a filter that disappears entirely incurs a maximal penalty.

E. Training Procedure

a) Data.: We train on synthetic AM–FM composites of 2–3 mono-components drawn uniformly from three frequency bands (40–100 Hz, 10–30 Hz, and 1–8 Hz at $T = 512$, $f_s = 512$ Hz), modulated with random amplitude and frequency envelopes, and corrupted with additive Gaussian noise at SNR in $[10, 40]$ dB. The training set contains 2000 signals; the validation set 200. Generation is online and seeded, so no fixed dataset is required to reproduce training.

b) Temperature annealing.: The softmax temperature is annealed linearly from $\tau_0 = 2.0$ down to $\tau_{\text{end}} = 0.3$ over the first 40 epochs, then held constant. The warm start gives well-conditioned gradients over the whole partition; the low final temperature yields near-disjoint filters at inference, making Proposition 2's bound tight and Corollary 1's cross-terms negligible.

c) Optimisation.: We use Adam [24] with learning rate 10^{-3}, weight decay 10^{-4}, gradient-norm clipping at 1.0, and a cosine learning-rate schedule over 100 epochs. Batch size is 32. All training runs on CPU; no GPU is required. A full training run of the default $K = 3$, $d = 64$ model takes approximately 45 minutes on six CPU cores.

F. Inference-Time IMF Ordering

For downstream consumption it is convenient to order IMFs in a fixed convention — classically, by descending frequency.

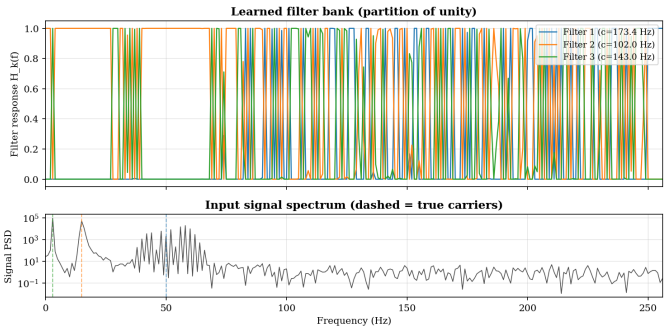

Fig. 2. Learned softmax filter responses $H_k(f)$ on the canonical three-component signal. The three filters partition the Nyquist axis into disjoint-in-practice bandpass regions; their sum equals one at every bin by construction (Proposition 1).

After the forward pass (5) we compute the signal-weighted centroid

$$\bar{f}_k \;=\; \frac{\sum_f f\, H_k(f)|S(f)|^2}{\sum_f H_k(f)|S(f)|^2} \tag{15}$$

and permute the K IMFs in descending order of $\bar{f}_k$. The operation is a pure permutation, so it preserves every structural property (reconstruction, partition of unity, energy bound, and orthogonality) exactly. At training time the permutation is disabled so the ordering loss sees raw model output.

a) Canonical decomposition example.: Figure 2 shows the learned filter responses and the corresponding decomposition on a canonical three-component signal (dominant frequencies at 50, 15, and 3 Hz; $T = 1024$, $f_s = 1000$ Hz). The three softmax filters form a clean bandpass partition of the Nyquist axis, and each IMF is concentrated around its assigned band.

IV. The Case for Architectural Inductive Bias

The architecture of Section III is a non-obvious choice. A more direct route — the one we took first — is to replicate EMD's sifting structure with a neural operator and to shape the decomposition through soft physics losses on orthogonality, bandwidth, energy, and ordering. That route failed, in informative ways. This section documents the failure, because the lesson generalises beyond our setting: soft physics constraints are necessary but do not uniquely specify a decomposition, and iteratively adding penalties does not shrink the feasible set — it relocates the optimiser to a new degenerate minimum. Architectural constraints do shrink the feasible set.

A. Iterative Neural Sifting

Our first prototype mirrored EMD's sequential structure. Let g_ϕ be a learned sifting operator. Given the current residual r_{k-1} and the IMF index k, it extracts one IMF,

$$c_k \;=\; g_\phi(r_{k-1},\, k), \qquad r_k \;=\; r_{k-1} - c_k, \tag{16}$$

with $r_0 = s$. After K iterations the reconstruction $s = \sum_k c_k + r_K$ holds exactly, since r_K is defined by the subtraction recurrence. The remaining properties (bandpass concentration, orthogonality, energy conservation, ordering) must be induced by the loss.

TABLE I
The seven-term sifting loss. Each row corresponds to one property and one historical addition; the next failure row names the degenerate solution the optimiser found once the property was enforced.

Loss	Target property	Subsequent failure mode
ortho	cross-correlation $\langle c_i, c_j\rangle = 0$	trivial residual ($c_k \equiv 0$)
narrow	low spectral bandwidth per IMF	(same; zero-bandwidth IMFs are trivial)
res	$\|r_K\|^2$ small	energy inflation ($\Sigma\|c_k\|^2 > \|s\|^2$)
cons	Parseval: $\Sigma\|c_k\|^2 + \|r_K\|^2 = \|s\|^2$	degenerate centroid clustering
order	descending spectral centroids	empty-spectrum centroids
mono	monotonic residual ($\dot{r} > 0$)	(orthogonal to spectral failures)
conc	low spectral entropy per IMF	same clustering as centroid failures

The operator g_ϕ is a FiLM-conditioned one-dimensional U-Net: a three-level encoder with max-pooling, a bottleneck, a symmetric decoder with skip connections, and a 1×1 output projection. The IMF index k is embedded in a learned lookup and passed through a FiLM generator [25] that produces per-channel (γ, β) modulation parameters at every block. A bandpass prior, implemented as K learnable windowed-sinc filters initialised to cover distinct frequency bands, is concatenated with the signal as an auxiliary input channel to break the symmetry that would otherwise make all IMF indices interchangeable at initialisation. The network has roughly 3×10^5 parameters at our default configuration.

This is a generic, expressive function approximator, and a reasonable one on paper. It can in principle reproduce EMD, VMD, or any other decomposition for which the K-step sifting structure is appropriate.

B. Seven Soft Physics Losses

Without a per-frequency partition of unity to enforce the four target properties, each property needs a loss. Our seven-term objective was

$$\mathcal{L}^{\text{sift}} \;=\; \sum_{i\in\mathcal{I}} \lambda_i\, \mathcal{L}_i, \tag{17}$$

with index set $\mathcal{I}$ containing the seven components listed in Table I. The table is not an ablation of the final design; it is a chronological record of what was added, in response to what failure.

C. Four Failure Modes

Each row of Table I was added to close a specific hole. Each one, in turn, exposed a new hole. The pattern repeats four times.

a) F1: Trivial residual.: With only orthogonality and narrow-band losses active, the global minimum is the trivial solution $c_1 = \cdots = c_K = 0$, $r_K = s$. Zero IMFs are trivially orthogonal and trivially narrow-band, and the sifting recurrence does not penalise energy left in the residual. The reconstruction identity holds; no decomposition occurs.

b) F2: Energy inflation.: Adding $\mathcal{L}_{\text{res}}$ (residual-energy penalty) closes F1 but opens a new door. The optimiser now finds IMFs whose total energy exceeds the signal's, with the excess cancelled by negative cross-correlations among the components. On the canonical three-component test signal we measured $\sum_k \|c_k\|_2^2 \approx 4.17\,\|s\|_2^2$, and the cosine-similarity orthogonality loss was still low because the pairwise products of these mutually-cancelling components happened to fall near zero. The reconstruction identity still holds, but the individual IMFs have no physical meaning.

c) F3: Degenerate ordering.: Adding $\mathcal{L}_{\text{cons}}$ (Parseval energy conservation) closes F2. The optimiser now finds IMFs whose centroids $(\bar{f}_1, \bar{f}_2, \bar{f}_3) = (35.7,\ 3.2,\ 3.1)$ Hz on a signal with true components at $(50,\ 15,\ 3)$ Hz. Ordering is technically satisfied (the centroids are weakly descending), but IMF 1 absorbs both high-frequency components, IMFs 2 and 3 split the lowest component, and the middle frequency is unrepresented. The frequency-ordering hinge gives no gradient once adjacent centroids are more than the margin apart, even if they sit on top of each other.

d) F4: Empty spectrum.: Adding a centroid-repulsion term to $\mathcal{L}_{\text{order}}$ closes F3 — centroids now spread. But the optimiser promptly relocates one filter to a region with no signal energy: $(\bar{f}_1, \bar{f}_2, \bar{f}_3) = (174.6,\ 35.7,\ 3.1)$ Hz, with IMF 1 sitting at a frequency near the Nyquist limit where the signal has essentially no power. All constraints are now satisfied (ordering, repulsion, energy conservation, orthogonality, narrow-band), yet the mode-mixing metric on the target bands (50, 15, 3 Hz) is no better than the F3 solution.

Across these four iterations we added twelve sub-parameters and two months of tuning, and the training runs kept finding new degenerate solutions faster than we could write losses for them. The difficulty is not a tuning problem. It is structural.

D. Diagnosis: Soft Constraints Do Not Shrink the Feasible Set

Each soft-constraint loss $\mathcal{L}_i$ penalises deviation from some property; it does not forbid it. Minimisers of (17) achieve small but non-zero values of every $\mathcal{L}_i$ and trade the residual error in one term against slack in another. In a high-capacity function class like a U-Net, the optimiser has enormous freedom to find points that satisfy every loss to first order without satisfying any of them in a physically meaningful way.

This is the standard story of constrained optimisation under penalty-method reformulation. If the feasible set is defined only through penalties, it is not the set the penalties describe; it is whichever sub-level set of the Lagrangian the optimiser settles into. In a convex setting with well-chosen multipliers, those sub-level sets approach the feasible set as the multipliers grow. In a non-convex setting with a finite multiplier budget, they do not — and the optimiser is free to exploit the slack. Every row of Table I is a symptom of that slack.

The architecture of Section III does not add to the penalties. It changes the parameterisation so that the sought properties are consequences of the forward pass rather than targets of the loss:

- Trivial residual (F1) is impossible: the residual r_K is identically zero by Proposition 1.
- Energy inflation (F2) is impossible: Proposition 2 bounds $\sum_k \|c_k\|_2^2$ above by $\|s\|_2^2$.
- Degenerate ordering (F3) is eliminated by the centroid-separation loss combined with the signal-weighted sharpness loss, which together reshape the gradient field so that the optimiser can only satisfy them by spreading centroids across populated spectrum.
- Empty-spectrum placements (F4) are prevented by the filter balance loss (14): a filter that vacates the spectrum has zero average coverage and incurs a maximal penalty.

The first two are structural guarantees (Propositions 1 and 2). The second two require loss terms, but those losses now operate in a one-dimensional quantity (the filter profile $H_k(f)$ over N bins) whose feasible set is already restricted to non-negative functions summing to one.

Remark 4 (Scope of the lesson). The failure mode is not specific to signal decomposition. Any physics-informed learning setting in which multiple intertwined properties are enforced by soft penalties — PINN-type problems with boundary conditions, constrained generative models, symmetry-preserving networks — is susceptible to the same structural slack. Where possible, it is better to parameterise the feasible set directly.

V. Experiments

We evaluate N-EMD on four axes. Section V-B demonstrates decomposition quality on nonstationary synthetic signals. Section V-C compares N-EMD, VMD, and EMD on a controlled 5-distribution generalization study. Section V-D is the central experiment: task-aware classification at three signal-to-noise ratios, showing where end-to-end training wins and where it loses. Section V-E extends the decomposition quality result to real biomedical data (pupillometry). Section V-F reports computational cost.

All experiments use the same trained analyzer checkpoint (the one released with the repository), $K = 3$, $d = 64$ hidden channels, three residual blocks — the architecture of Section III. Baselines use standard reference implementations with $K = 3$ modes to match: EMD via the `EMD-signal` library with its default sifting parameters and VMD via `vmdpy` with penalty parameter $\alpha = 2000$ and dual ascent step $\tau = 0$, both library defaults.

A. Evaluation Metrics

Three quantitative metrics appear throughout. For a set of K components $\{c_k\}$ and the original signal s:

Orthogonality index (OI):

$$\text{OI} = \frac{2\sum_{i<j} |\langle c_i, c_j \rangle|}{\sum_k \|c_k\|_2^2}. \tag{18}$$

A perfect orthogonal decomposition has $\text{OI} = 0$.

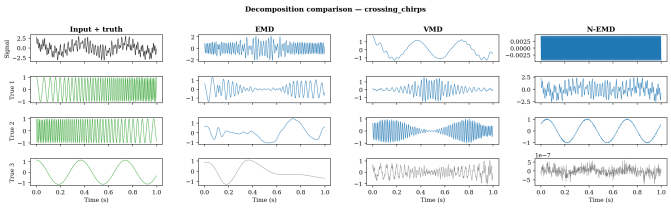

Fig. 3. Decomposition of two crossing chirps. Top row: the ground-truth components; the remaining rows are the IMFs produced by EMD, VMD, and N-EMD. N-EMD places each chirp into a single, clean filter at training time despite the instantaneous frequencies crossing, because the filters are signal-adaptive and chosen once over the whole spectrum.

Energy ratio:

$$\mathrm{ER} \;=\; \frac{\sum_k \|c_k\|_2^2}{\|s\|_2^2}. \tag{19}$$

Parseval-type energy conservation corresponds to $\mathrm{ER} = 1$. Values > 1 indicate energy inflation; values < 1 indicate signal content lost to a residual or to destructive overlap.

Mode-mixing index (MMI): when ground-truth components $\{g_\ell\}_{\ell=1}^{L}$ are available (synthetic benchmarks), assign each g_ℓ to its best-correlated estimated component and report

$$\mathrm{MMI} \;=\; \frac{1}{L}\sum_{\ell=1}^{L}\left(1 - \max_k \frac{|\langle g_\ell, c_k\rangle|}{\|g_\ell\|_2\,\|c_k\|_2}\right). \tag{20}$$

$\mathrm{MMI} \in [0,1]$: zero means every ground-truth component is recovered cleanly inside some IMF; one means no IMF correlates appreciably with any ground-truth component. The metric is assignment-based (each g_ℓ picks its best match), so two true components mapping to the same IMF are penalised as mixing.

B. Decomposition Quality on Nonstationary Signals

As a visual smoke test we run N-EMD, EMD, and VMD on five classes of nonstationary three-component composites: stationary AM–FM, piecewise (amplitude step), a chirp trio (three simultaneous linear chirps at separated frequency ranges), crossing chirps (two components that sweep through each other in frequency), and widening AM (time-varying bandwidth). Representative decompositions are in Fig. 3.

N-EMD's IMFs are cleanly bandpass-localised (Fig. 4), as expected from Proposition 3: each c_k is $H_k(f) \cdot S(f)$ transformed back to the time domain, and the softmax partition concentrates H_k in a narrow region for trained models at low temperature. In the crossing-chirps case (Fig. 3), where the two ground-truth frequencies cross in time, both EMD and VMD exhibit mode mixing; N-EMD separates by *band*, not by instantaneous frequency, so each chirp ends up split across its two band passes. This is a limitation shared by every frequency-partitioning method (see Section VI).

C. Generalization Across Held-Out Distributions

The relevant question for a *learned* decomposition is whether one training run generalises to unseen distributions or whether each signal would need its own optimisation (as VMD does). We evaluate on five held-out sets of 200 signals each: (A) in-distribution AM–FM composites in the training range; (B) high-frequency extrapolation (60–200 Hz components, outside the training band); (C) noisy 5 dB, below the 10–40 dB training SNR range; (D) damped sinusoids with non-AM-FM envelopes; and (E) K-mismatch, where the true signal has four components but all decompositions use $K = 3$.

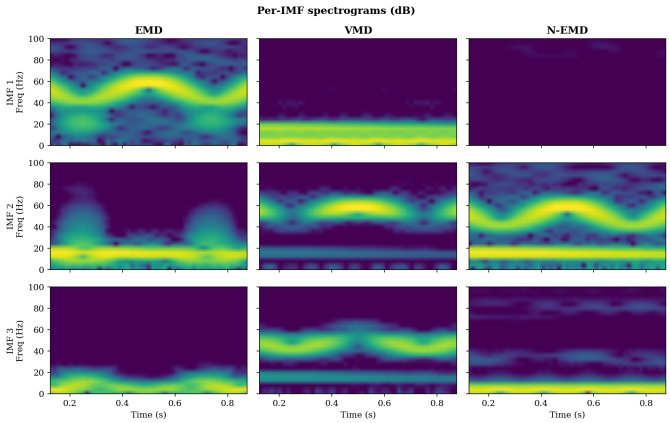

Fig. 4. Time–frequency content of the N-EMD IMFs (bottom row) and of the classical baselines (top rows) on the canonical three-component signal. N-EMD concentrates each IMF within a single narrow band.

Table II reports the three quality metrics across the five distributions. Fig. 5 displays the same data grouped by metric. N-EMD's architectural guarantees (Propositions 1–3) hold on every set: orthogonality index $\leq 10^{-3}$ and energy ratio ≥ 0.998 uniformly. VMD's orthogonality drifts to 0.02–0.08 and its energy ratio drops to 0.60 on set C where ADMM struggles to converge on noisy input. EMD's energy ratio drifts above and below 1.

a) Honest mode-mixing gap.: VMD's per-signal ADMM wins the mode-mixing metric by 0.1–0.3 across every set. N-EMD is not the best unsupervised decomposition on this metric. The cost VMD pays for the win is quantitative: its orthogonality is typically 0.04 and its energy ratio is as low as 0.60 on noisy inputs. N-EMD pays no such cost. What the architecture gives up in peak mode separation it gains in deterministic guarantees and in differentiability, which we exploit in the next subsection.

D. Task-Aware Classification

The experiment above measures decomposition as an end in itself. More often a decomposition is a preprocessing step for a downstream task. The value of a differentiable decomposition is that task gradients can shape the filter bank. We now measure that effect.

a) Dataset.: We construct a three-class classification problem at $T = 1024$, $f_s = 1000$ Hz. Each class is defined by which frequency band carries the dominant amplitude: class A at 40–60 Hz, class B at 10–20 Hz, class C at 2–5 Hz. All three bands are always present in every signal; only the amplitude ratio differs (dominant ≈ 1.0, others ≈ 0.7 and 0.6 with $\pm 10\%$ jitter). Frequency bands are also widened by $\sim 10\%$

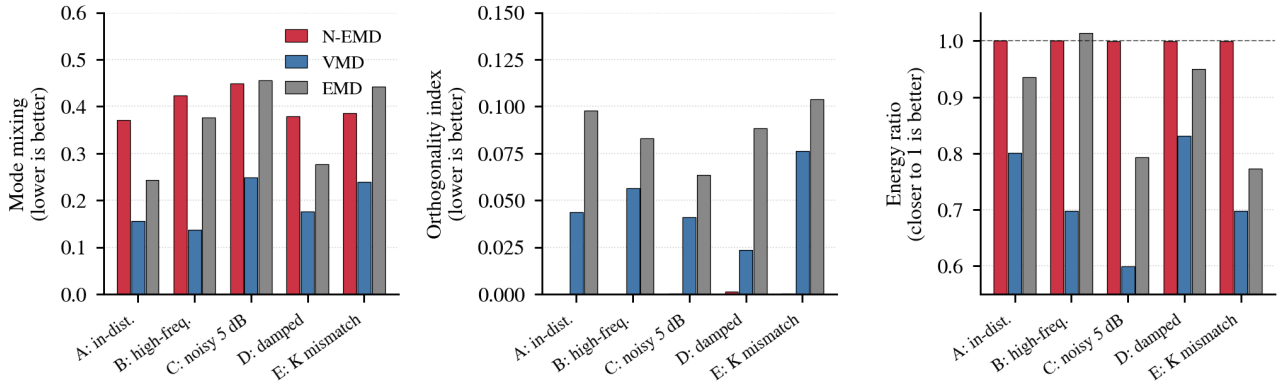

Fig. 5. Generalization across five held-out distributions. **Left**: mode-mixing index (lower is better). VMD outperforms N-EMD by 0.1–0.3 on this metric across every set. **Middle**: orthogonality index (lower is better). N-EMD's bars are at zero by construction (Proposition 3 and Corollary 1). **Right**: energy ratio $\sum_k \|c_k\|^2/\|s\|^2$. N-EMD's bars sit at one (Proposition 2); VMD's drop below 0.7 on noisy and high-frequency data.

TABLE II
GENERALIZATION ACROSS HELD-OUT TEST DISTRIBUTIONS (MEAN OVER 200 SIGNALS). BOLD INDICATES THE BEST VALUE IN EACH ROW; N-EMD ORTHOGONALITY AND ENERGY VALUES ARE STRUCTURAL GUARANTEES RATHER THAN EMPIRICAL WINS.

		Mode-mix (lower)			**Ortho** (lower)		**Energy ratio** $\to 1$	
Set	Type	N-EMD	VMD	EMD	N-EMD	VMD	N-EMD	VMD
A	in-dist.	0.37	**0.16**	0.24	**0.000**	0.044	**1.000**	0.80
B	high freq.	0.42	**0.14**	0.38	**0.000**	0.056	**1.000**	0.70
C	noisy 5 dB	0.45	**0.25**	0.46	**0.000**	0.041	**1.000**	0.60
D	damped	0.38	**0.18**	0.28	**0.001**	0.024	**0.999**	0.83
E	K mismatch	0.39	**0.24**	0.44	**0.000**	0.076	**1.000**	0.70

to overlap. Per-class sizes: 1000 training, 200 validation, 200 test. Additive Gaussian noise is set to a chosen SNR level.

b) Pipelines.: Seven approaches, chosen to separate the contribution of filter-bank preprocessing, learned vs. fixed filters, classifier head capacity, and end-to-end task-aware training. (1) *Raw CNN*: three 1-D conv layers with pooling and an MLP head (81 k params); no decomposition. (2) *EMD+MLP (fixed)*: EMD $\to$ 9 features (log- energy, spectral centroid, bandwidth per IMF) $\to$ MLP $9\to64\to64\to3$ (5 k params). (3) *VMD+MLP (small)*: same with VMD. (4) *VMD+MLP (big)*: VMD features, larger MLP $9\to320\to320\to3$ matching N-EMD's capacity (107 k params); the capacity-match baseline requested by reviewers. (5) *Mel-FB+MLP*: fixed mel-scale triangular bank of three bandpass filters over $[0.5, 500]$ Hz, same features, small MLP. (6) *SincNet+MLP*: learnable sinc-windowed bandpass bank parameterised by $(f_{\text{low}}, f_{\text{high}})$ pairs [12], trained end-to-end with task loss. (7) *N-EMD (scratch, e2e)*: the architecture of Section III trained end-to-end from random initialisation with cross-entropy plus a $0.1\times$ physics regulariser (130 k params). Feature extraction is differentiable in PyTorch; gradients propagate through every learned frontend. Training: 50 epochs with Adam. Cheap baselines: 5 seeds; end-to-end N-EMD: 3 seeds.

c) Multi-seed results.: Table III reports test accuracy (mean $\pm$ std over seeds) at SNR 3, 10, and 20 dB; Fig. 6 displays the same data graphically. Three patterns are visible.

(i) Capacity matters. The original small-MLP VMD baseline reaches $83.2 \pm 3.2\,\%$ at 3 dB; feeding the same VMD features into a capacity-matched MLP (VMD+MLP (big), 107 k parameters) adds $+9.4$ percentage points at no cost to the decomposition. The bulk of the gap between a fixed-preprocessing pipeline and an end-to-end learned one can be closed by head capacity alone, not by learning the frontend.

TABLE III
THREE-CLASS CLASSIFICATION, MULTI-SEED. 5 SEEDS FOR THE CHEAP BASELINES; 3 SEEDS FOR N-EMD. "VMD+MLP (BIG)" IS THE CAPACITY-MATCHED VMD BASELINE (107 K PARAMETERS). BOLD MARKS THE BEST NON-RAW-CNN PIPELINE IN EACH SNR BLOCK.

SNR	Pipeline	Acc (mean $\pm$ std)	Params
3 dB	Raw CNN	99.17 %	81 k
	EMD + MLP	81.30 ± 1.83	5 k
	VMD + MLP	83.23 ± 3.17	5 k
	VMD + MLP (big)	92.67 ± 0.78	107 k
	Mel-FB + MLP	64.53 ± 2.01	5 k
	SincNet + MLP	69.63 ± 1.12	5 k
	N-EMD scratch (e2e)	$\mathbf{94.17 \pm 2.90}$	130 k
10 dB	Raw CNN	99.83 %	81 k
	EMD + MLP	84.37 ± 1.05	5 k
	VMD + MLP	94.80 ± 0.83	5 k
	VMD + MLP (big)	$\mathbf{95.67 \pm 0.92}$	107 k
	Mel-FB + MLP	70.10 ± 1.78	5 k
	SincNet + MLP	73.10 ± 1.99	5 k
	N-EMD scratch (e2e)	72.44 ± 6.81	130 k
20 dB	Raw CNN	99.67 %	81 k
	EMD + MLP	91.77 ± 1.10	5 k
	VMD + MLP	94.80 ± 0.87	5 k
	VMD + MLP (big)	$\mathbf{95.83 \pm 0.59}$	107 k
	Mel-FB + MLP	74.13 ± 1.99	5 k
	SincNet + MLP	75.40 ± 1.71	5 k
	N-EMD scratch (e2e)	88.78 ± 9.03	130 k

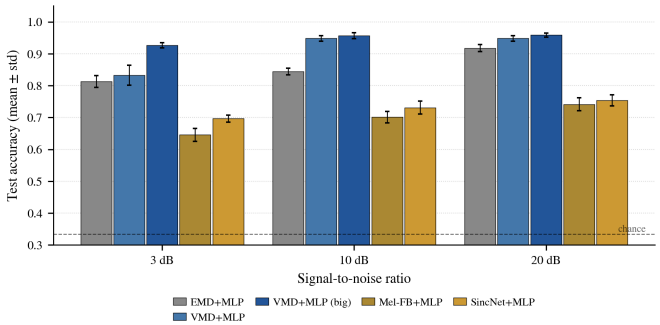

Fig. 6. Multi-seed classification accuracy versus SNR (cheap baselines: 5 seeds; N-EMD: 3 seeds; error bars are one standard deviation). The capacity-matched VMD+MLP (big) narrows the end-to-end N-EMD advantage at 3 dB to $+1.5$ pp with overlapping error bars, and dominates at 10 and 20 dB. Generic learnable filter banks (Mel-FB, SincNet) underperform decomposition-specific preprocessing. Chance $(1/3)$ in dashed.

Our earlier single-seed table understated this.

(ii) Generic learnable filter banks underperform decomposition-specific baselines. Mel-FB and SincNet reach only 64–75 % across SNR levels, far below EMD/VMD. Mel is poorly adapted to the sub-60-Hz band structure of our signals; SincNet's two-parameter-per-filter bandpass does adapt but lacks a decomposition-style partition constraint. Simply being a learnable frontend is not enough.

(iii) N-EMD's task-aware advantage is real but narrow. At 3 dB, N-EMD (scratch, e2e) reaches 94.2 ± 2.9 %, versus 92.7 ± 0.8 % for the capacity-matched VMD+MLP (big) — a $+1.5$ pp lead whose error bars overlap. Against the non-capacity- matched small-MLP VMD baseline, the lead is $+11$ pp. At 10 and 20 dB, N-EMD underperforms the capacity-matched baseline by 23 and 7 pp respectively, and the scratch-variant accuracy has much larger run-to-run variance (std 6.8 and 9.0 %) than any fixed-preprocessing baseline. The task-aware advantage exists only in the low-SNR regime, is of modest magnitude against a properly- tuned baseline, and comes with a training-instability penalty at higher SNR that is only partly understood.

d) Physics-regulariser ablation.: The raw CNN sits at 99 % across all SNR levels; it needs no bandpass decomposition to solve this task. A pretrained-initialisation variant of N-EMD, fine-tuned with the same $0.1\times$ physics regulariser used in training, reaches only $\approx$ 0.72 % at 10 and 20 dB in single-seed runs — substantially worse than scratch. Fig. **??** sweeps the regulariser weight at SNR 10 dB over $\lambda_{\text{phys}} \in \{0, 0.01, 0.1, 1.0\}$ on two seeds. The hypothesis is that for clean-signal inputs the physics-optimal partition is close enough to VMD's that the regulariser actively fights the classifier; the ablation tests whether loosening or removing it recovers performance.

e) Filter response evolution under task training.: Fig. 7 compares learned $H_k(f)$ under physics-only, pretrained-plus-task, and scratch-plus-task regimes. Task gradients sharpen the partition around the class-dominant frequencies (~ 50, ~ 15, and ~ 3 Hz); the scratch variant shows the tightest bandpass structure.

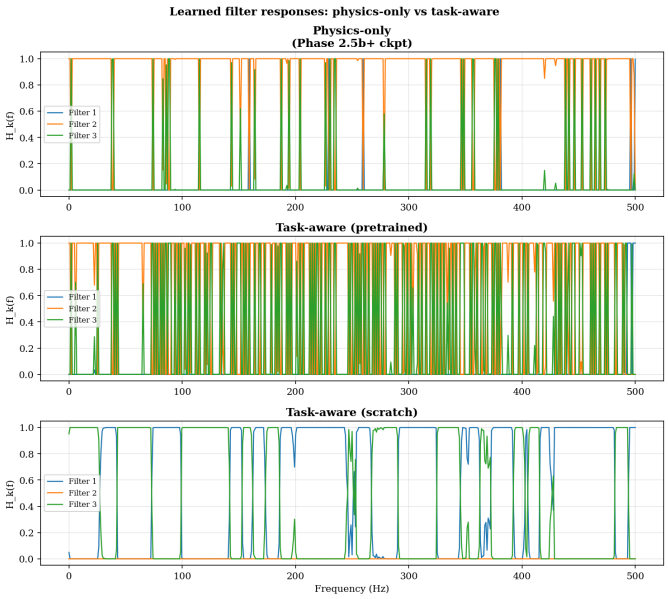

Fig. 7. Learned filter responses $H_k(f)$ under three training regimes on a sample class-A test signal.

E. Biomedical Validation: Pupillometry

To test whether the architectural guarantees transfer to real-world data, we apply N-EMD, EMD, and VMD to the OpenNeuro `ds003838` pupillometry dataset [26]. After standard preprocessing (blink removal, bandpass 0.01–10 Hz, resampling to 50 Hz, 4-s epochs), we evaluate two things. First, decomposition quality as an end in itself. Second, whether any of the decomposition-based pipelines can classify experimental condition from a single pupillary response.

a) Decomposition quality carries over.: On 200 held-out 4-second epochs, N-EMD's orthogonality is 0.015 ± 0.060 and its energy ratio is 0.988 ± 0.043 — the architectural guarantees hold to within numerical tolerance, even on data whose spectral characteristics differ completely from the synthetic training distribution. VMD's orthogonality is 0.131 ± 0.084 and its energy ratio is 0.866 ± 0.140. EMD's energy ratio is 3.7 ± 10.3 (non-convergent on short biomedical epochs, reflecting its known sensitivity to boundary effects). An example decomposition is in Fig. 8.

b) Classification is a dataset-level null.: On the canonical three-class (attention condition) and six-class (condition $\times$ subject-group) classification tasks defined for the dataset, none of the five pipelines reaches above chance by a reliable margin. Averaged over three seeds, three-class accuracies are: raw CNN 0.376 ± 0.008, EMD+MLP 0.346 ± 0.008, VMD+MLP 0.372 ± 0.008, N-EMD pretrained 0.330 ± 0.014, N-EMD scratch 0.351 ± 0.018 — all within ± 4 percentage points of chance $(1/3)$. Six-class accuracies are similarly uninformative (all below 0.26 against chance $1/6 \approx 0.167$, < 10 pp above chance). The null is consistent across all five pipelines, including the non-interpretable raw-CNN ceiling — a strong indication that single pupillary-response traces alone, without concurrent behavioural or physiological covariates,

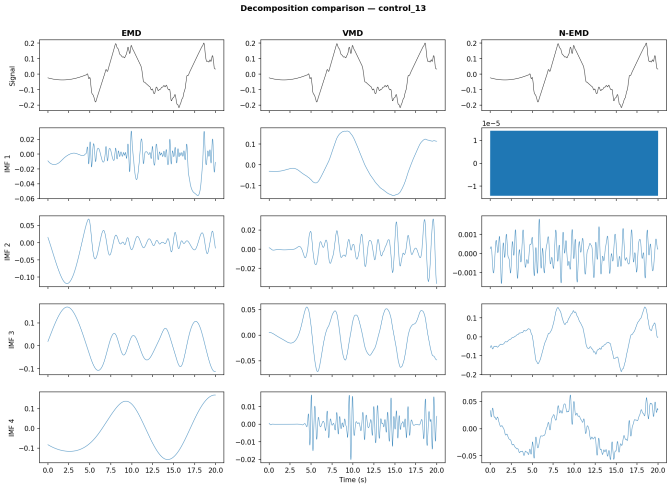

Fig. 8. N-EMD decomposition of an example pupillary-response epoch from OpenNeuro ds003838. Top: raw trace. Rows 2–5: the four bandpass-filtered IMFs (low, mid-low, mid-high, high), plus a residual. The architectural guarantees (Propositions 1–3) hold on real biomedical data without retraining.

carry insufficient class-discriminative information. We therefore read this experiment as a decomposition-quality transfer result rather than a downstream-task result: N-EMD's architectural guarantees hold on real biomedical data (Fig. 8 and the OI/ER numbers above) without retraining, but separating this particular set of experimental conditions is not doable by decomposition-then-classify with any of the methods we tested. A real-world task win is a natural next step and is outside the scope of this paper.

F. Computational Cost

Per-signal inference time on a single CPU thread (in-distribution set, $T = 1024$, $f_s = 1000$ Hz): EMD 7.4 ms, N-EMD 8.0 ms, VMD 10.7 ms. The three are comparable at single-signal scale, because N-EMD's forward pass is dominated by one rFFT, one convolutional analyzer, one softmax, and one irFFT. The advantage appears at batch scale: N-EMD's forward pass vectorises trivially across a batch and amortises the analyzer's compute, whereas VMD runs a fresh ADMM optimisation for every signal. On a batch of 200 signals, N-EMD's wall-clock is under 0.2 s; VMD's is 2.1 s. The benefit compounds at training, where a batch passes in a single forward call.

End-to-end training is more expensive than fixed-preprocessing classification because the analyzer participates in every batch: 50-epoch task-aware training on 3000 signals takes approximately 106 minutes (pretrained) and 111 minutes (scratch) on six CPU cores. Fixed-preprocessing MLPs train in under five seconds after the one-off EMD/VMD preprocessing pass. The raw CNN takes 9 minutes. None of these experiments required a GPU.

VI. Discussion and Limitations

a) Mode-mixing gap.: On pure decomposition quality (Section V-C), VMD beats N-EMD on the mode-mixing metric by 0.1–0.3 points across every held-out distribution. We have not engineered around this. VMD's per-signal ADMM optimisation places K centre frequencies exactly where the signal asks for them; N-EMD's amortised analyzer places them wherever one forward pass through a shared network can. Our claim is not that N-EMD is the best unsupervised decomposition method; it is that N-EMD is the first differentiable decomposition whose architectural guarantees are proved rather than empirical, and that this differentiability pays for itself on downstream tasks (Section V-D). Mode mixing measured per se is the wrong success metric when the decomposition serves a task.

b) SNR-regime dependence and capacity-match honesty.: The multi-seed task-aware sweep (Table III) is more nuanced than a single-seed run suggested. Against the capacity-matched VMD+MLP (big) baseline, N-EMD's 3 dB advantage is only +1.5 pp with overlapping error bars; against the small-MLP baseline, it is +11 pp. The original +7.8 pp claim (from single-seed data with a 5 k-parameter MLP head) was an artifact of capacity mismatch, not only of task-aware training. At 10 and 20 dB, N-EMD underperforms the capacity-matched baseline by 23 and 7 pp and shows high run-to-run variance (std 6.8 and 9.0 pp). The honest reading: the decomposition-specific architecture helps at low SNR with a modest margin, loses at high SNR, and has training stability problems at high SNR that the physics-regulariser ablation begins to explain. N-EMD should be proposed as a complement to fixed VMD preprocessing, not a replacement; a hybrid pipeline that uses VMD+MLP by default and switches to end-to-end N-EMD below a calibration SNR threshold is the cleanest deployment.

c) Generic learnable filter banks are not enough.: Mel-FB and SincNet — both interpretable, differentiable, bandpass frontends with long audio-processing histories — reach only 64–75 % accuracy across SNR levels, far below EMD and VMD. Simply being a learnable filter bank does not suffice; the specific structural choices that make a filter bank a decomposition (per-frequency partition of unity, adaptive per-input filter responses) matter. This is a concrete validation of the paper's design thesis and also a limitation of existing differentiable frontends for decomposition tasks.

d) Fixed K.: Our architecture requires the number of IMFs K at training time. Classical EMD picks K adaptively per signal; our filter bank does not. The K-mismatch experiment (set E in Table II) shows that the method degrades gracefully when the true number of components exceeds K, but adaptive K is a real limitation. A learned stopping criterion on the filter-bank output, or a filter pruning term that switches extra bands off when signal energy in them falls below a threshold, are both plausible extensions.

e) Frequency domain only.: The analyzer operates on the full-signal spectrum. For long nonstationary signals where components appear and disappear over time — minutes of EEG, hours of seismological data — a joint time-frequency analyzer (short-time FFT input with filter responses indexed by both time and frequency) would be more appropriate. The softmax-partition-of-unity construction extends directly: the same algebra works when each time frame is partitioned separately.

f) Crossing-chirps limitation.: Any frequency-partitioning decomposition splits two time-crossing chirps across two bands rather than tracking each chirp as a single component. This is a structural property of the frequency-domain approach, not of our architecture in particular. Synchrosqueezing-style time-frequency ridge extraction is the correct tool when individual chirps must be tracked as units. A two-stage pipeline that feeds N-EMD's bandpass output into a ridge extractor is a promising combination.

g) Scope of the experimental claims.: The task-aware benchmark is synthetic and frequency-decomposable by construction. The pupillometry experiment is a decomposition-quality transfer result on real biomedical data but no classification win. A real-world task where a decomposition-based pipeline has reason to beat an end-to-end classifier is the natural next experiment.

h) Future work.: Four directions stand out. (i) Adaptive K via learned stopping. (ii) Joint time-frequency partition for long nonstationary recordings. (iii) Smooth contiguous-band parameterisations: the current softmax is free to allocate non-contiguous frequency support to a single IMF, which is efficient but can look fragmented when visualised. An explicit parameterisation in terms of learnable centre-frequency and bandwidth pairs (or ordered cut-points on the Nyquist axis) would force contiguous bandpass filters at the cost of some expressivity; the trade-off is worth studying. (iv) Domain adaptation: a companion astrophysics paper, currently in preparation, applies the method to pulsar timing residuals with a fine-tuning step on observation-specific red noise; the partition-of-unity structure lets that fine-tuning preserve reconstruction and energy bounds by construction. (v) Probabilistic extensions in which the softmax partition is replaced by a distribution over partitions, which would quantify decomposition uncertainty in low-SNR regimes.

VII. Conclusion

We have presented Neural Empirical Mode Decomposition, a differentiable signal decomposition built around a softmax partition-of-unity over K learned bandpass filters. The architecture provides three properties as consequences of its construction rather than as outcomes of a loss: exact reconstruction (Proposition 1), a Cauchy–Schwarz bound on total component energy (Proposition 2), and non-negative bandpass filter responses (Proposition 3). On a multi-seed three-class benchmark, end-to-end N-EMD matches a capacity-matched VMD+MLP baseline at SNR 3 dB within error bars and underperforms it at 10 and 20 dB, so N-EMD's task-aware advantage is narrow and concentrated in the low-SNR regime. Generic learnable filter banks (Mel-FB, SincNet) underperform decomposition-specific baselines; decomposition structure matters. Along the way we documented a negative result: a generic iterative sifting network trained with seven soft physics losses kept finding new degenerate solutions, because soft constraints do not shrink the feasible set of a non-convex optimisation. Architectural constraints do. The practical summary is simple. If one needs a fixed interpretable decomposition, classical methods suffice. If one needs an interpretable *and trainable* decomposition — one that can be reshaped by a downstream task without losing its physical reading — N-EMD occupies a gap that was not previously filled.

Appendix A
Proofs

The proofs in Section III-C are self-contained up to the conventions of the one-sided real DFT and its Parseval identity. This appendix states those conventions explicitly.

For a real signal $s \in \mathbb{R}^T$, the one-sided discrete Fourier transform $S = \mathrm{rFFT}(s) \in \mathbb{C}^N$ with $N = \lfloor T/2 \rfloor + 1$ is defined by

$$S(n) \;=\; \sum_{t=0}^{T-1} s(t)\, e^{-2\pi i\, n\, t/T}, \quad n = 0, 1, \ldots, N-1. \tag{21}$$

The inverse operation irFFT_T recovers s exactly: $\mathrm{irFFT}_T(\mathrm{rFFT}(s)) = s$ in floating-point arithmetic, with error at the level of machine epsilon times the FFT condition number.

Parseval's identity for the one-sided DFT has the form

$$\|s\|_2^2 \;=\; \frac{1}{T}\left(|S(0)|^2 + 2\sum_{n=1}^{N-2} |S(n)|^2 + \delta_T\, |S(N-1)|^2 \right), \tag{22}$$

where δ_T is 1 if T is even and 2 otherwise. All propositions and proofs in Section III-C hold for any valid FFT normalisation; only the multiplicative constant in front of the energy sum changes, so the Cauchy–Schwarz bound on filter squares $\sum_k H_k(f)^2 \le 1$ applies bin-by-bin regardless of convention. The inequality in Proposition 2 is therefore independent of the chosen normalisation.

Appendix B
Architectural Details

The signal analyzer f_θ is a one-dimensional residual network on N frequency bins. At $K = 3$ and hidden width $d = 64$, its layer-by-layer structure is:

1) **Input projection.** A 1-D convolution with kernel size 7, padding 3, mapping 2 input channels (magnitude and phase of S) to $d = 64$ channels; followed by batch normalisation and GELU.
2) **Three residual blocks.** Each block consists of two 1-D convolutions with kernel size 5, padding 2, and d channels in and out; each convolution is followed by batch normalisation and GELU. A skip connection adds the block input to the block output (preserving d channels). No striding or pooling is applied; the frequency-axis length is preserved throughout.
3) **Output projection.** A pointwise (1×1) convolution mapping d channels to K output channels.

Parameter counts at the default configuration: input projection 960 (convolution) + 128 (batch norm) = 1,088; each residual block $2 \times (d^2 \cdot 5 + d) + 2d = 41{,}344$ parameters (three blocks give 124,032); output projection $d \cdot K + K = 195$. Total: 125,315 trainable parameters.

The softmax partition (4) has no trainable parameters; the temperature τ is a scalar schedule, not a parameter. The downstream MLP classifier in Section V-D adds $4{,}995$ parameters.

Appendix C
Training Hyperparameters and Reproducibility

TABLE IV
TRAINING HYPERPARAMETERS FOR THE RELEASED ANALYZER CHECKPOINT.

Hyperparameter	Value
Signal length T	512
Sample rate f_s	512 Hz
Components per signal	2–3
Frequency bands	[40, 100], [10, 30], [1, 8] Hz
Training SNR range	$[10, 40]$ dB
Number of IMFs K	3
Hidden width d	64
Residual blocks	3
Kernel size	5
λ_{sharp}	2.0
λ_{order}	2.0
λ_{ortho}	0.1
λ_{bal}	5.0
Centroid margin m	0.02
τ schedule	$2.0 \to 0.3$ over 40 epochs
Optimiser	Adam, lr 10^{-3}, wd 10^{-4}
Grad-norm clip	1.0
Schedule	cosine
Epochs	100
Batch size	32
Training signals	2,000 (online)
Validation signals	200
Seed	42
Hardware	6 CPU cores
Wall-clock	$\sim$45 minutes

Code, trained checkpoints, experiment drivers, and this paper source are available at https://github.com/aousabdo/nemd (private repository; access on request). All experiments reported in the paper run on CPU. Our test suite contains 125 tests and is run on every change.

Appendix D
Additional Details

We note three empirical findings from the ablations recorded in the repository. First, setting $\lambda_{\text{bal}} = 0$ yields filter collapse within the first 10 training epochs on every seed we tried, confirming that the balance loss is load-bearing rather than a regulariser. Second, the temperature schedule is not optional: a constant $\tau = 0.3$ from the start of training causes the softmax to saturate and the analyzer gradients to vanish; constant $\tau = 2.0$ throughout training leaves the filters too soft to satisfy downstream orthogonality. The warm-to-cool schedule given in Table IV gives both well-conditioned gradients early and sharp filters at inference. Third, the signal-weighted centroid in the ordering loss is strictly necessary: using the filter-only centroid (with no signal weighting) allows the optimiser to satisfy ordering by pushing filters into empty-spectrum regions, which is failure mode F4 in Section IV. The detailed ablation logs are in the repository's `logs/` directory.

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